



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	MECHANICS AND OPTICS				
Type of Course	Minor (GROUP I: MATHEMATICS FOR PHYSICAL SYSTEMS)				
Semester	I				
Academic Level	100 – 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamentals of vectors, calculus and kinematics.				
Course Summary	This course explores Newton's Laws of Motion and how they can be applied to solve different mechanical systems, and also discusses various phenomena exhibited by light.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply Newton's Laws of Motion to solve different mechanical systems	Ap	P	Instructor-created exams / Home Assignments
CO2	Apply work-energy theorem to solve different mechanical systems	Ap	P	Instructor-created exams / Home Assignments
CO3	Analyse conservative systems and solve them using the conservation of mechanical energy.	An	P	Instructor-created exams / Home Assignments
CO4	Understand the basic nature and different phenomena exhibited by light.	U	C	Instructor-created exams / Home Assignments

CO5	Develop a skill to analyse the behaviour of light beams in devices consisting of mirrors and lenses.	Ap	P	Seminar Presentation / Group Tutorial Work
CO6	Develop skills to set up and perform experiments to test Newton's Laws of Motion, work energy theorem and different phenomenon exhibited by light.	Ap & C	P	Practical Assignment / Observation of Practical Skills / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	NEWTON'S LAWS OF MOTION AND APPLICATIONS		12	19
	1	Newton's first laws: particles in equilibrium, Inertial frames of reference	3	
	2	Newton's Second law: Dynamics of particles	3	
	3	Frictional forces	3	
	4	Dynamics of circular motion	2	
	5	Fundamental forces of nature	1	
	Sections 4.2, 5.1 – 5.5 of Book1			
II	WORK AND ENERGY		11	17
	6	Work, Kinetic energy and work energy theorem	3	
	7	Work and energy with varying forces	3	
	8	Gravitational potential energy	2	
	9	Elastic potential energy	1	

	10	Conservative and non-conservative forces	1	
	11	Force and potential energy	1	
	Sections 6.1- 6.3, 7.1 - 7.4 of Book 1			
III	GEOMETRICAL OPTICS		11	17
	12	Nature of light, reflection, refraction	2	
	13	Total internal reflection, Dispersion	2	
	14	Reflection and refraction at a plane surface, reflection at spherical surface	3	
	15	Refraction at a spherical surface	2	
	16	Thin lenses, camera	2	
	Sections 33.1 - 33.4 of chapter 33 and sections 34.1 - 34.5 of chapter 34 of Book 1			
IV	INTERFERENCE AND DIFFRACTION		11	17
	17	Interference and coherent source	1	
	18	Two source interference of light, intensity of interference pattern	3	
	19	Interference in thin films, Newtons rings	1	
	20	Diffraction, Fresnel and Fraunhofer diffraction	1	
	21	Single slit diffraction	3	
	22	Two slits, Multiple slits	2	
	Sections 35.1 - 35.4 of chapter 35 and sections 36.1- 36.4 of chapter 36 of book 1			
V	PRACTICALS		30	
	Conduct any 5 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 6 th experiment may also be selected from the given list. Other experiments			

	listed here may be used as demonstrations of the concepts taught in the course. - Plot the graphs using GeoGebra. FitLine function may be used to get the slope. - Smartphones are exclusively intended for educational lab use. Necessary care should be taken to safeguard them during the experiments. - Smartphone experiments primarily serve demonstration purposes, with result accuracy contingent upon the precision of phone sensors and experimental setups.		
1	Coefficient of Static Friction. - Determine the coefficient of static friction between a wooden block and a wooden plane. - Measure the angle at which the wooden block just starts to slide down an inclined wooden plane and hence calculate the static friction coefficient. https://www.youtube.com/watch?v=gt8mr6pFSFE OR - Place the wooden block on a wooden plane surface and add mass to the pan attached to the block using a string through a frictionless pulley. - Find the mass required to initiate the sliding of the block. - Different trials can be done by adding mass on the top of the block and hence determine the coefficient of static friction. - Example 5.13 of Book 1. https://www.youtube.com/watch?v=MSV6VafiUF4&t=443s		
2	Verification of Newton's First Law: Equilibrium of a Particle - Analyze the two dimensional equilibrium problems using spring/digital force gauges. - Hang a weight from a chain that is linked at the ring to two other chains, one fastened to the ceiling and the other to the wall. Example 5.3 of Book 1. - Measure the angle between the chain from the ceiling and the horizontal and the tension in each of the three chains using		

		spring/digital force gauges and verify with the theoretical predictions. • https://www.youtube.com/watch?v=XI7E32BROp0		
	3	Acceleration of a Freely Falling Body • Use the smartphone acoustic stopwatch to determine the duration of a free fall. • Measure the time of flight of a steel ball for different heights and plot a graph of distance vs. time squared (s vs. t^2). Determine g from the graph. • Experiment 2 of Book 2. • Phyphox app may be used. https://phyphox.org/experiment/free-fall-2/ OR • Use ExpEyes kit, electromagnet, and contact sensor to determine the duration of a free fall. https://expeyes.in/experiments/mechanics/tof.html		
	4	Verification of the Relation of Angular Velocity and Centrifugal Acceleration • Use the smartphone gyroscope and the accelerometer. • Attach the smartphone to some rotating arrangements and record the data from the gyroscope and accelerometer. • Plot angular velocity Vs acceleration and verify the relation. • Experiment 18 of Book 2. • https://doi.org/10.1119/1.4872422 • Phyphox app may be used. https://phyphox.org/experiment/centrifugal-acceleration/		
	5	Analysis of Air Resistance and Terminal Speed to Determine the Drag Coefficient. • Record the motion of a light weight paper cup and analyse it with Tracker tool (https://physlets.org/tracker/). • Plot acceleration, velocity, and position with time. • Repeat the experiment with different mass (by simply stacking the paper cups)		

		- Determine the Drag Coefficient - Experiment 27 of Book 2. https://www.youtube.com/watch?v=ijzK3uH1Yc		
	6	Projectile Motion: Energy Conservation - Analyse the motion of the tossing ball/ projectile in the Tracker tool. - Plot time vs the x-and y-components of velocity and acceleration. - Also plot the kinetic energy, potential energy (build data using define tool) and total energy. https://www.youtube.com/watch?v=x0AWRLvgB28 https://www.youtube.com/watch?v=i07HeUWo8xc		
	7	Analysis of Bouncing Balls to Determine Gravitational Acceleration and Coefficient of Restitution. - After doing the experiment, the student should be able to understand the concept of inelastic collision. - Measure the time interval between successive bounces using a digital acoustic stopwatch and hence calculate g and coefficient of restitution - Experiment 12 of Book 2 and section 3.3 of Book 1 - Phyphox app may be used. https://phyphox.org/experiment/inelastic-collision/		
	8	The Nearly Parabolic Trajectories of a Bouncing Ball - Perform Experiment 7 using Tracker tool. - Track the ball and plot the time vs position graph. - Measure the time interval between successive bounces and hence calculate g and coefficient of restitution. - Experiment 12 of Book 2 and section 3.3 of Book 1 - Tracker Autotracker Tutorial: https://www.youtube.com/watch?v=Dn0Zz7rtkZw		
	9	Determine the refractive index of (a) given liquid and (b) the material of a lens, by forming a liquid lens. Through this experiment the students are expected to get the concepts of image formation, combination of lenses and radius of curvature of the surface of lens.		

		Determine the radius of curvature of the lens by Boy's method and hence calculate the refractive indices.		
	10	Determine the focal length of the combination of two lenses separated by a distance. - Determine the focal lengths, f_1 and f_2 of the two lenses using an illuminated cross-slit screen holder, nodal slide(for placing the lenses) and plane mirror arrangement. - Place the two lenses separated by a distance d, determine the focal length, F of the combination and verify the relation $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$. - The combination of the lenses in the eyepiece of the spectrometer/ travelling microscope may be used for the study. https://www.youtube.com/watch?v=IOIEEtyNPBg https://www.youtube.com/watch?v=tNo4Ipk74SU		
	11	Determination of the refractive index of the material of the prism - Familiarize the initial adjustments and measurements in the spectrometer. - Find the angle of the prism and the angle of minimum deviation using the yellow line of a sodium lamp and calculate the refractive index.		
	12	Determination of the dispersive power of a solid prism using a spectrometer. - Find the angle of the prism and the angle of minimum deviation for prominent lines of the mercury spectrum using a spectrometer. - Calculate the refractive indices corresponding to the colors and find the dispersive power of the material of the prism for two pairs of wavelengths.		
	13	Determination of wavelengths of mercury spectrum using diffraction grating and spectrometer. - Arrange the grating at normal incidence. - Standardize the grating using the green line of mercury and then find the wavelengths of other prominent lines of the spectrum.		

	14	Newton's rings-determination of the wavelength of sodium light - Form of Newton's rings in the air-film in between a plano-convex lens and a glass plate using sodium-source. - Determine the radius of curvature by Boy's method and determine the wavelength of the source.		
	15	Air wedge-determination of the radius of a thin wire/human hair//thin foil. - Form interference fringes using sodium-source, in the air-film in between wedge formed by placing the given sample between the glass plates. - Measure the positions of the successive dark bands using a travelling microscope and determine the angle of the wedge and thickness of the sample given.		
	16	Single slit diffraction using laser - Determination of slit width. - The laser light diffracted from the narrow slit is allowed to fall on a screen and record the maxima or minima points in a paper. - From the width of the central maxima or the position of minimum intensity points, calculate the slit width. - Wavelength of laser can be found using diffraction grating of known N.		

Books and References:

- University Physics with Modern Physics (Edn.15) by Young & Freedman (Book 1)
- Smartphones as Mobile Minilabs in Physics(Edn. 1) by Jochen Kuhn & Patrik Vogt, Springer, (Book 2)
- <https://phyphox.org/>
- <https://physlets.org/tracker/>
- Berkeley Physics Course : Vol.1 : Mechanics, 2ndEdn. – Kittel et al. – McGraw-Hill
- Optics by Ajoy Ghatak – 4th edition
- A textbook of Optics by Subramaniam, Brijlal & Avadhanulu, 25th Edition- S Chand and Company Limited

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	1	2	0	2	2	2	0	0	2	2	0
CO 2	2	2	1	2	0	2	2	2	0	0	2	2	0
CO 3	2	2	2	2	0	2	2	2	0	0	2	2	0
CO 4	0	1	0	1	2	1	2	2	0	0	2	2	0
CO 5	0	0	0	0	2	0	2	2	0	0	2	2	0
CO 6	2	2	2	2	0	2	2	2	0	0	2	2	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	